Cavity Resonators

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Lecture Objective

By the end of this lecture, you will be able to:

- Understand the basics of cavity resonators.
- Evaluate resonant frequency of modes supported by a rectangular cavity.
- Evaluate resonant frequency and Q of dominant TE101 mode of rectangular cavity.

Cavity Resonators

- A hollow conducting box (“cavity resonator”) is often used as a resonant circuit to provide selectivity at microwave frequencies.
- One way to construct a cavity resonator is by closing the end faces of a section of hollow-tube waveguide.
- These cavity resonators replace lumped RLC circuits at frequencies where the parasitics associated with the latter degrade performance dramatically.
- Cavity resonators are used for frequency control of oscillators as well as for frequency selectivity of filters and amplifiers.

Rectangular Cavity Geometry

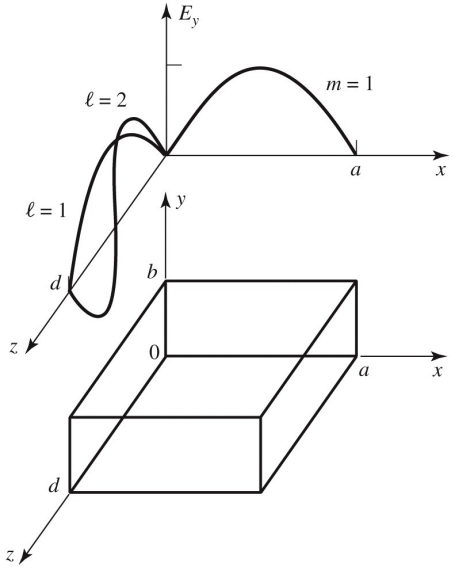

Figure 6.6
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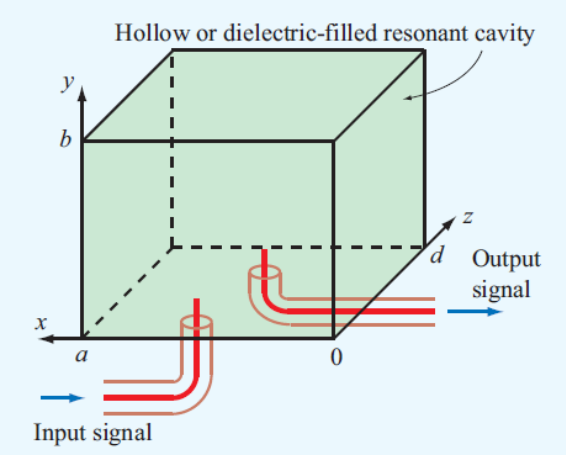
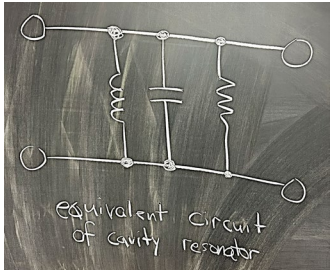
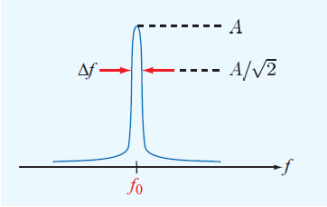
- The rectangular cavity resonator can be viewed as a section of rectangular waveguide with both ends closed by conducting walls.
- By convention, we arrange the parallelepiped such that $a > d > b$.

Coupling Energy Into/Out of a Cavity Resonator







Frequency response

Fields in Cavity

- The z-variation is no longer $\exp(-\gamma z)$ because of the walls at $z=0$ and $z=d$.
- The fields in all three directions can be treated as the superposition of incident and reflected waves bouncing off the cavity walls.
- Any mode supported by a homogeneously filled rectangular cavity will be either TE or TM.

Nomenclature for TM and TE Modes

- We designate the TM and TE modes of the rectangular cavity as TM_{mnp} and TE_{mnp} respectively.
- The indices m, n, and p specify the variation in x-, y- and z-directions respectively.
- For TM modes: m and n are both positive integers; p is a non-negative integer.
- For TE modes: m and n are both non-negative integers but they cannot both be zero; p is a positive integer.

Resonant Frequencies of Cavity Modes

$$f_{mnp} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

Where we have assumed a cavity filled homogeneously with a non-magnetic dielectric medium, most often air.

- The ‘dominant’ mode of a waveguide cavity is the mode having the lowest resonant frequency.
- The dominant mode of a rectangular cavity will be the mode with the smallest value of the quantity

(over the permitted sets of integer values for TM and TE modes)

$$\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2$$

Dominant Mode of Rectangular Cavity

- Assuming the convention that $a > d > b$, we find that the dominant mode is the lowest order TE mode (TE_{101}) with

$$f_{101} = \frac{c}{2\sqrt{\varepsilon_r}} \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{d}\right)^2}$$

Questions

Consider a cavity formed from WR90 waveguide of length a $d = 0.6$ inch length of WR90 waveguide.

What is the resonant frequency of the TE101 mode?

What is the mode with the next lowest resonant frequency?

Dominant Mode of Rectangular Cavity

- The electric and magnetic fields are given by

$$\tilde{H}_z = \frac{j\pi E_0}{\omega\mu_0}\cos\left(\frac{\pi x}{a}\right)\sin\left(\frac{\pi z}{d}\right)$$

Eigenfunction derived from DE and BCs

$$\tilde{H}_x = \frac{-jE_0}{Z_{TE}}\sin\left(\frac{\pi x}{a}\right)\cos\left(\frac{\pi z}{d}\right)$$

$$\tilde{E}_y = E_0\sin\left(\frac{\pi x}{a}\right)\sin\left(\frac{\pi z}{d}\right)$$

Here, we followed Pozar's definition of the leading constant

Energy Dissipated in Cavity Resonator

- Energy loss in a cavity results from
  - Dielectric loss
  - Conductor loss
  - Coupling in/out of cavity

Unloaded Q

Loaded Q

Quality Factor of Cavity Resonator

- The "***quality factor***" – usually just called "Q" – of an electrical, acoustic or mechanical resonator is defined as

energy stored in resonator at resonance

$$Q = 2\pi \frac{W_S}{W_D} = \frac{\omega W_S}{P_L}$$

energy dissipated in resonator over one cycle at resonance

Quality Factor of Cavity Resonator

- For the TE_{101} mode, assuming an air-filled cavity (and thus no dielectric losses):

$$Q_{101} = \frac{\pi f_{101} \mu_0 abd \left(a^2 + d^2\right)}{R_s \left[2b\left(a^3 + d^3\right) + ad\left(a^2 + d^2\right)\right]}$$

$$R_s = \sqrt{\frac{\pi f \mu_c}{\sigma_c}}$$

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